Concept A fallback kit · page 1 · scale check + turret + rod plates

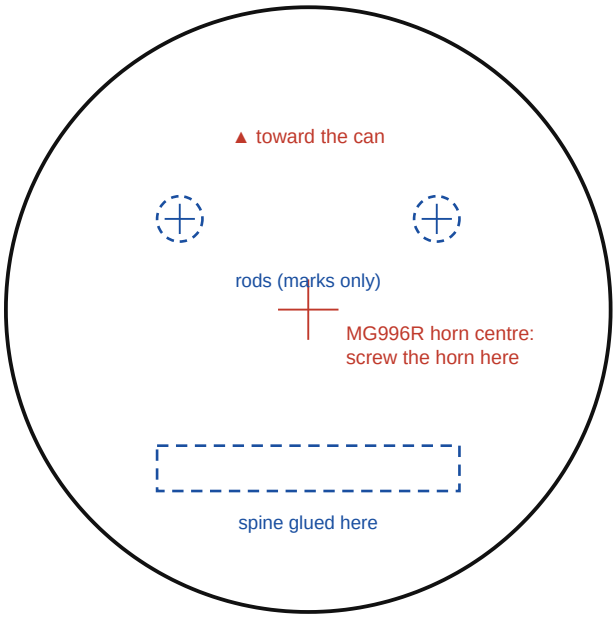
Solid line = cut. Red cross = hole centre. Dashed = glue / position mark. Foam board 5 mm.

50 mm square

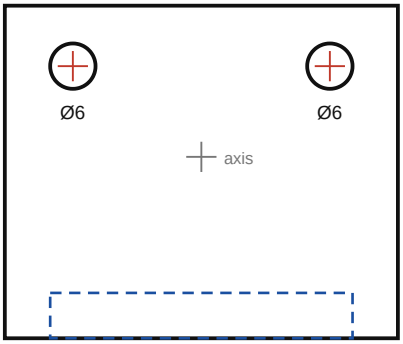
Measure it. Not 50 mm? Printer scaled the page: reprint at 100 %.

2 in = 50.8 mm

TURRET DISC r 40 mm (cut 2, glue together)

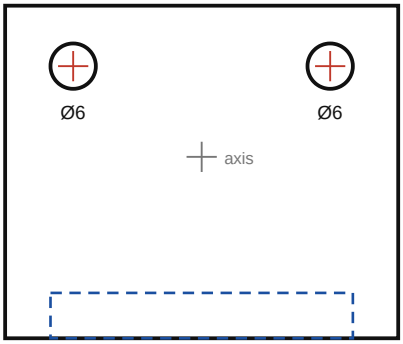


ROD PLATE, BOTTOM (cut 2, glue)



spine edge here · rods 34 mm apart

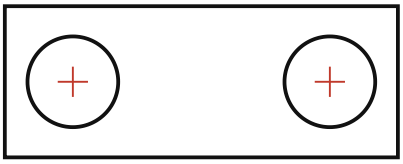
ROD PLATE, TOP (cut 2, glue)



spine edge here · rods 34 mm apart

Rods: push through the bottom plate, hot-glue, plates 112 mm apart (outer faces). Bottom plate glued onto the turret disc with its axis cross on the disc's cross. Holes snug: start Ø5, open with the rod itself.

CARRIAGE ENDS (cut 2)



Ø12 holes for the LM6UU. Join the 2 ends with foam-board sides 62 mm tall. LM6UU hot-glued in, + a zip tie each.

LIFT LINK: holes 11.4 mm apart



popsicle stick, drill Ø3.5 (#6 screw pivot)

CRANK: horn centre → link hole = 20 mm



lay the lift servo's long horn on this, mark, drill Ø1.5 for a servo screw

PRINT AT 100 % ("Actual size", not "Fit"). Check the 50 mm square on page 1 with a ruler.

Concept A fallback kit · page 2 · gauges + spine

Cut out the strips and use them while gluing. Heights are measured from the TABLE.

REACH GAUGE: turntable axis → can centre = 127 mm (5 in)

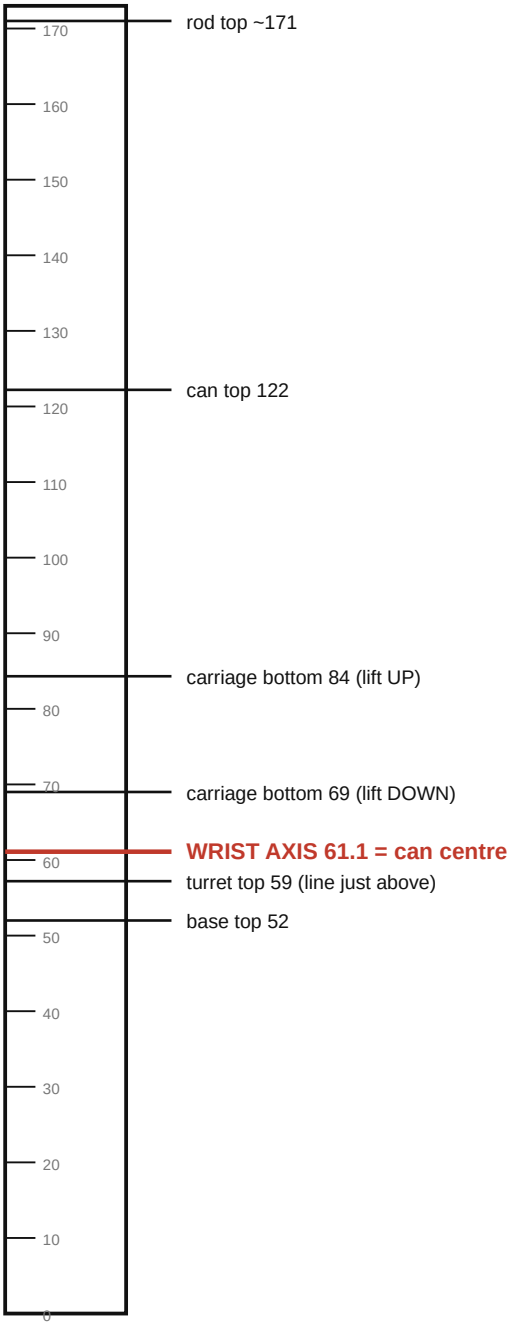


Pin the left cross on the turntable axis; the right cross must land on the can's centre when the claw holds it.

ARM: carriage front → wrist servo face = 36.5 mm

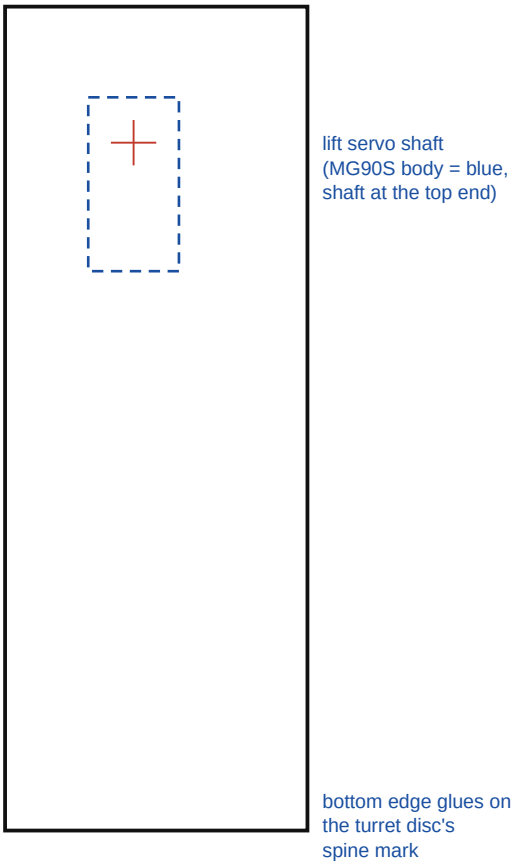


2 popsicle sticks glued as a sandwich, wrist servo hot-glued at the right end, its shaft pointing at the can, shaft centre at the 61.1 mark when the lift is DOWN.



HEIGHT GAUGE (stand it on the table)

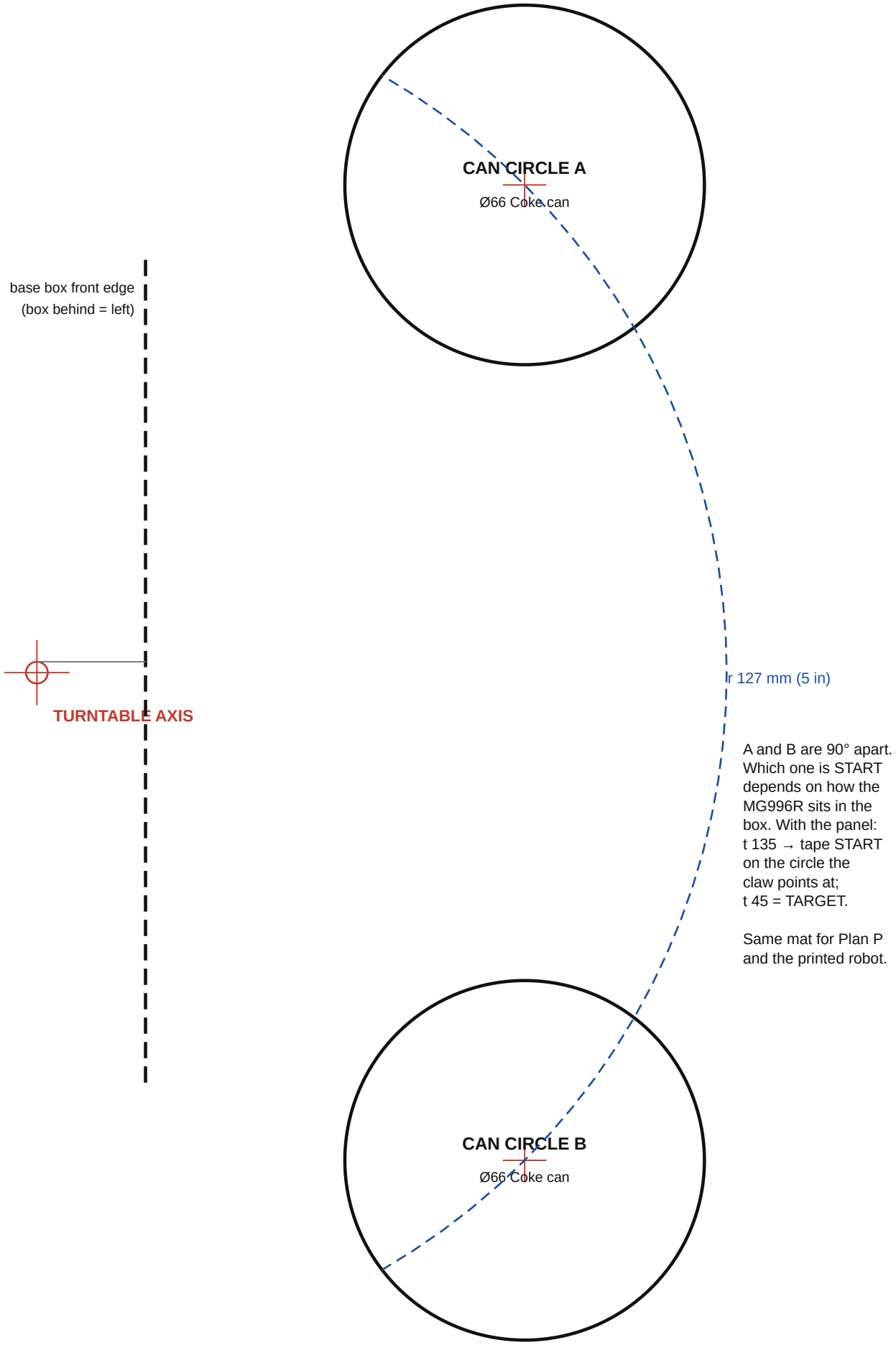
SPINE (cut 2, glue) 40 × 109 mm



PRINT AT 100 % ("Actual size", not "Fit"). Check the 50 mm square on page 1 with a ruler.

Concept A fallback kit · page 3 · floor mat (turntable axis, START and TARGET circles)

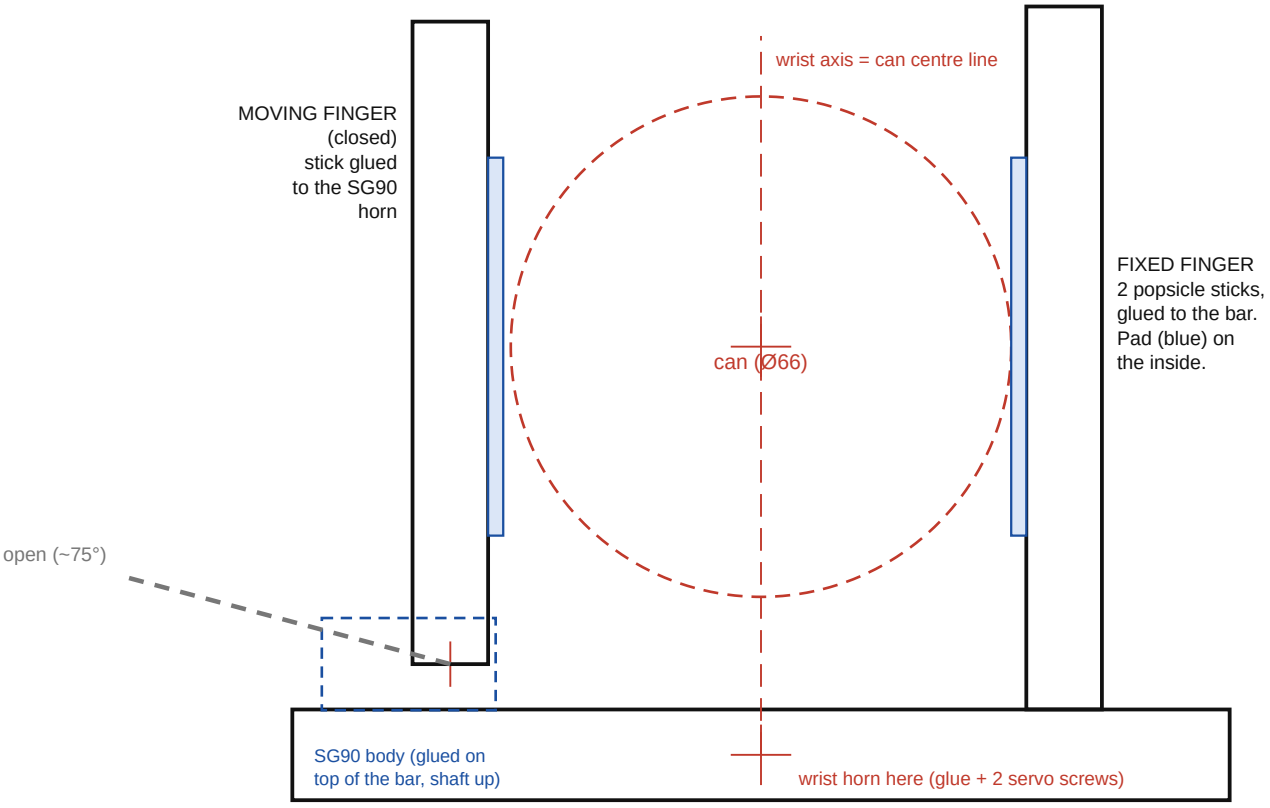
Tape this to the table. Robot axis on the cross; base box behind the line.



PRINT AT 100 % ("Actual size", not "Fit"). Check the 50 mm square on page 1 with a ruler.

Concept A fallback kit · page 4 · no-print claw (only if nothing can be printed)

Top view, 1:1. One moving finger (SG90 horn + stick), one fixed finger. Pads: rubber band / inner tube, ~2 mm.



CROSS BAR: foam board 124 × 12 (or 2 popsicle sticks)

The can sits between the pads, its centre on the wrist axis (red line). Closed: pads squeeze the can ~1 mm each side. The SG90 angles will differ from the printed claw: set GRIP OPEN / CLOSED / RELEASE in the control panel's calibration table, then Copy code into concept_a.ino.